Cartier–Miller evaluation of genus-one coefficient sums

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Abstract

Let $p \geq 7$ be prime and let $f \in \mathbb{F}_p[w]$ be squarefree of degree three or four with $f(0) = 1$. We evaluate the truncation $T_f = f^{(p-1)/2} \bmod w^p$ at every nonzero $\mu \in \mathbb{F}_p$ with $f(\mu) \neq 0$ through an elliptic trace and a normalized logarithmic derivative. The formula includes curves of trace ± 1, where the curve or its quadratic twist is anomalous. For trace 1 the exceptional normalization assembles classical results of Deuring, Cassels, Gunji and Voloch; we prove the required residue-divisor form for both signs directly in characteristic p, with the scalar fixed explicitly. A differentiated Miller recurrence gives an $O(\log p)$ field-operation query after trace preparation; combined with Schoof's algorithm, this yields deterministic polynomial time in $\log p$. Computation in a quadratic algebra avoids a nonresidue search. The original quarter-point weighted central-binomial sum is a worked application whose boundary coefficient is classical complex-multiplication data. A separate rounded-third extension identifies its additional boundary as a normalized de Rham entry and supplies checked batch and single-prime backends; it does not extend the polynomial-time claim to that boundary.

Introduction and main theorem

Consider evaluation at a base-field point of the degree-below-p truncation of a polynomial power. Put $h=(p-1)/2$ and write

$$f(w)^h=\sum_m c_m\, w^m, \qquad T_f(w)=\sum_{m=0}^{p-1} c_m\, w^m, \qquad H=c_{p-1}. \tag{1}$$

For $\lambda\neq 0$ in $\mathbb{F}_p$, the weighted coefficient expression is exactly this truncation evaluated at $\mu=1/\lambda$:

$$S_f(\lambda)=[w^{p-1}]\frac{f(w)^h}{1-\lambda w}=\sum_{m=0}^{p-1} c_m\, \lambda^{p-1-m}=T_f(1/\lambda). \tag{2}$$

The last equality uses $\lambda^{p-1}=1$ and is a base-field identity. At $\lambda=0$, the coefficient expression gives $S_f(0)=H$; evaluation $T_f(0)=1$ is separately immediate. Euler's criterion also gives $T_f(\mu)=\chi-\sum_{m\geq p} c_m\, \mu^m$, where $\chi=(f(\mu)/p)$ when $f(\mu)\neq 0$.

Let E be the smooth projective curve $v^2=f(w)$ with origin $O=P_+=(0,1)$, invariant differential $\omega=dw/v$, and exact trace $\tau=p+1-\#E(\mathbb{F}_p)$. We call $\lambda\in\mathbb{F}_p$ admissible if $\lambda=0$, or if $\lambda\neq 0$ and $f(1/\lambda)\neq 0$. For a nonzero admissible argument define

$$\begin{gathered} d=f(1/\lambda), \quad v_0^2=d, \quad \chi=(d/p), \\ Q_\pm=(1/\lambda,\pm v_0), \quad D=(Q_+)-(Q_-), \quad r=\frac{1}{\lambda v_0}. \end{gathered} \tag{3}$$

For an integer M annihilating the class of D, fix a rational function F_D with $\operatorname{div}(F_D)=MD$, and set $\Psi_M(D)=\big((dF_D/F_D)/\omega\big)(O)$. This value is independent of the scalar multiplying F_D. For $p\nmid M$ it retains the specified divisor representative; for $M=p$ it depends only on its divisor class.

Theorem 1 (Uniform base-field evaluation). Under these hypotheses, $H=\tau \bmod p$ and every nonzero admissible argument satisfies

$$S_f(\lambda)=H-\chi r\Psi_{M_\chi}(D), \qquad M_\chi=p+1-\chi\tau. \tag{4}$$

The product $r\Psi_{M_\chi}(D)$ belongs to $\mathbb{F}_p$ and is unchanged by replacing v_0 by $-v_0$. The formula includes $\tau=\chi$, when $M_\chi=p$, and requires no division by that annihilator. After preparing the curve and its trace, an admissible query costs $O(\log p)$ field operations. Corollary 4 supplies the deterministic polynomial-time setup.

The query bound excludes preparation of the exact trace: for one query on a new curve, point counting can dominate the total cost, while repeated queries on the same curve and prime reuse that preparation. The main

theorem concerns genus one, exponent $(p-1)/2$, coefficient cutoff $p-1$, values modulo p, non-branch evaluation arguments, and $f(0) \neq 0$ after the stated normalization.

The regular identity follows from cancellation of residues and Cartier semilinearity. Lemma 3 supplies the exceptional normalization by identifying a Cartier defect with the established characteristic-p additive transfer. The contributions considered here are the coefficient-sum identity, its explicit scalar for both signs, the trace -1 residue-divisor formulation, and the algorithmic consequence. The transfer, the differentiated Miller recurrence and the trace 1 identification of the Artin–Schreier quasi-period with a logarithmic differential are established (Gunji, 1976; Voloch, 1990, 1997, n.d.; Semaev, 1998). No priority claim is made for the underlying additive map.

Truncated binomial and hypergeometric congruences motivate the evaluation problem. The question is which large truncations can be evaluated through a small geometric object rather than accumulated term by term. Direct accumulation at a cutoff proportional to p takes $\Theta(p)$ field operations, while polynomial-block recurrence methods have approximately square-root cost up to logarithmic factors (Bostan et al., 2007). The result here fixes the coefficient cutoff at p-1 and varies the evaluation point. It does not supply a uniform polylogarithmic algorithm for arbitrary central-binomial stopping indices. The quarter-point application shows how its particular prefix and boundary coefficient recover an original weighted sum. The rounded-third subsection separates another fast prefix from its additional factorial-equivalent boundary and records the scope of the companion implementations.

Relation to prior work

Serre (1958, Propositions 8 and 10) records Cartier's properties and identifies geometric p-torsion divisor classes with Cartier-fixed holomorphic differentials through $d\log$. Semaev (1998, Lemmas 1–3) uses this additive transfer and a differentiated Miller recurrence for discrete logarithms on elliptic p-torsion. With our invariant differential fixed, his local evaluation $(F_D'/F_D)(R)$ is $\Psi_p(D)$ times the local coefficient of ω at R. Rück (1999) extends the construction to divisor class groups of curves.

For trace 1, the curve has exactly p rational points and is anomalous; for trace -1, its quadratic twist is anomalous. Satoh and Araki (1998, corrected in 1999) use elliptic Fermat quotients, and Smart (1999) uses a p-adic lift, to solve the anomalous-curve discrete-logarithm problem. These belong to the same established setting. Voloch (n.d., pp. 3–4) explicitly relates his descent map to Semaev's logarithmic transfer and compares it with the Smart and Satoh–Araki maps. Thus the identification of these maps is also prior work; the purpose here is to specify the coefficient-sum scalar relative to ω and the actual residue divisor.

In short Weierstrass coordinates, write $F(x)^{(p-1)/2} = \mathcal{U}(x) + Hx^{p-1} + x^p\mathcal{V}(x)$, with $\deg\mathcal{U} \leq p-2$. When the trace is 1, $H = 1$ and Verschiebung is zero on $E(\mathbb{F}_p)$, since $\pi + V = 1$ and π is the identity on rational points. Voloch's descent map therefore gives an injective homomorphism $E(\mathbb{F}_p) \to \mathbb{F}_p$, represented by $y\mathcal{V}(x)$ (Voloch, 1990, Proposition 1.3; n.d., p. 3). Once its coefficients have been prepared, evaluating this polynomial by Horner's rule takes $O(p)$ field operations; this statement does not include expanding $F^{(p-1)/2}$. The established logarithmic-transfer recurrence evaluates the same normalized map in $O(\log p)$ operations (Semaev, 1998).

For trace 1 the exceptional normalization is classical in substance. The high-part Artin–Schreier construction goes back to Deuring (1941; see Voloch, 1990, Lemma 1.1, and 1997, Section 2). Cassels (1971, Sections 1, 5–6) writes the cover as $g^p - Hg = y\mathcal{V}(x)$ and relates translations by kernel points to additive quasi-periods. Gunji (1976) studies the logarithmic differentials of functions with divisor $p((P)-(O))$ and their Cartier invariance. Voloch (1990, Lemma 1.1 and Proposition 1.3) identifies the cover with Verschiebung and obtains the descent homomorphism. Voloch (1997, Proposition 1 and the discussion after Proposition 3) develops its quasi-periodicity and its relation to logarithmic derivatives; he also corrects the differential scale and the sign convention of the 1990 construction.

Lemma 3 assembles these ingredients over a finite field. It fixes the scalar relative to $\omega = dw/v$ and the actual residue divisor, and includes trace -1, where a factor 2 enters. We have not located this negative-eigenvalue residue-divisor statement, or the two-pole base-field form of Theorem 1, in the sources considered here; that observation does not establish novelty. Frobenius acting on third-kind differentials also occurs in Coleman–

Gross height computations (Balakrishnan & Besser, 2012), and Cartier actions on polynomial-power coefficients appear in Dwork crystals (Beukers & Vlasenko, 2021). These are adjacent frameworks.

There is also a distinct many-primes workload. Harvey and Sutherland (2014) compute Hasse–Witt matrices modulo all good primes below a bound in average polynomial time. Their amortization across primes differs from the present preparation of one curve over one field followed by point-evaluation queries. The rounded-third companion specializes this many-primes approach to a triangular recurrence; its amortized workload is distinct from both a single-prime query and repeated queries over one prepared field.

Cartier evaluation and the regular identity

All differential identities are in characteristic p. Frobenius π on points means the p-power map on coordinates. Coefficient conjugation σ raises constants to their pth powers while leaving coordinate variables fixed. Put

$$\omega=\frac{dw}{v}, \qquad \theta_\lambda=\frac{dw}{(1-\lambda w)v}. \tag{5}$$

The differential ω is holomorphic and nowhere zero. For nonzero admissible λ, the only poles of θ_λ are simple poles at Q_+ and Q_- with residues $-r$ and r.

Let $\mathcal{C}$ denote absolute Cartier, with local convention

$$\mathcal{C}\left(\sum_n b_n z^n\, dz\right)=\sum_j b_{pj+p-1}^{1/p} z^j\, dz. \tag{6}$$

It satisfies $\mathcal{C}(a^p\eta)=a\mathcal{C}(\eta)$ and $\mathcal{C}(dF/F)=dF/F$ (Cartier, 1957; Serre, 1958, Proposition 8). Since $f(0)=1$, w is a uniformiser at O. The relation $v^{p-1}=f^h$ gives the coefficient selection directly:

$$\begin{aligned} \theta_\lambda&=v^{-p}\frac{f^h}{1-\lambda w}\,dw, \\ \mathcal{C}(\omega)=H\omega, \qquad &\left(\frac{\mathcal{C}(\theta_\lambda)}{\omega}\right)(O)=S_f(\lambda)^{1/p}. \end{aligned} \tag{7}$$

Here $H\in\mathbb{F}_p$, so $H^{1/p}=H$. The trace relation is the genus-one Hasse–Witt identity (Silverman, 2009, Chapter V), and can also be checked by the power sums

$$\sum_{w\in\mathbb{F}_p} w^k=\begin{cases}-1, & k>0 \text{ and } p-1\mid k,\\ 0, & \text{otherwise,}\end{cases} \qquad \text{in } \mathbb{F}_p. \tag{8}$$

For a cubic, only c_{p-1} survives in the sum of $f(w)^h$, and the projective curve has one point at infinity. For a quartic, $c_{2p-2}=f_4^h=(f_4/p)$ also survives; the infinity contribution is $1+(f_4/p)$ and cancels this extra term. In both cases $\tau\equiv H \pmod p$.

Proposition 2 (Regular residue identity). Let $\lambda\neq 0$ be in any perfect extension of $\mathbb{F}_p$, with $f(1/\lambda)\neq 0$. Suppose $M\geq 1$, $p\nmid M$, and fix F_D with $\operatorname{div}(F_D)=MD$ as divisors. Then

$$M\big(S_f(\lambda)-H\big)=r\left(H\Psi_M(D)-\Psi_M(D)^p\right). \tag{9}$$

Proof. The residues cancel in $M\theta_\lambda+r\,dF_D/F_D$, so this differential is $c\omega$ for some scalar c. Evaluation at O gives $c=M+r\Psi_M(D)$. Apply Cartier, evaluate at O, and raise to the pth power to obtain

$$MS_f(\lambda)+r\Psi_M(D)^p=\big(M+r\Psi_M(D)\big)H. \tag{10}$$

This is (9). For the fixed divisor D, replacing F_D by F_D^k gives $\Psi_{kM}(D)=k\Psi_M(D)$ in characteristic p.

For a base-field argument, $\pi(Q_+)=Q_\chi$, so $\pi(D)=\chi D$ as divisors. Frobenius's characteristic equation annihilates its divisor class by $M_\chi=p+1-\chi\tau$. Since F_D^σ differs from F_D^χ by a constant, $\Psi_{M_\chi}(D)^p=\chi\Psi_{M_\chi}(D)$. Substitution into (9), using $M_\chi\equiv 1-\chi H \pmod p$, proves (4) when $p\nmid M_\chi$. Changing the sign of v_0 negates both r and $\Psi_{M_\chi}(D)$.

Hasse's bound gives $|\tau|\leq 2\sqrt{p}<p-1$ for $p\geq 7$. Thus $p\mid M_\chi$ occurs precisely when $\tau=\chi$, and then $M_\chi=p$. The class of D is nonzero: distinct points on a genus-one curve cannot have principal difference. It therefore has order p. The regular identity loses its scalar in this case. The bound excludes $p=5$, where traces ± 4 are possible, from this particular degeneracy argument.

Direct exceptional normalization

Assume $\tau=\chi\in\{1,-1\}$ and identify $E=(v^2=f(w),P_+)$, so $O=P_+$ and $\omega=dw/v$. Work over $k=\overline{\mathbb{F}}_p$. For a degree-zero divisor D_0 with p-torsion class, choose $\mathrm{div}(F_{D_0})=pD_0$. Its logarithmic differential is holomorphic and defines

$$dF_{D_0}/F_{D_0}=\Psi_p([D_0])\omega, \qquad \Psi_p:\mathrm{Pic}^0(E)[p](k)\to k. \tag{11}$$

A principal change $D_0\mapsto D_0+\mathrm{div}(G)$ changes the function by a constant times G^p and leaves its logarithmic differential unchanged. Multiplication of functions gives additivity. The map is injective by Serre's logarithmic-differential construction (Serre, 1958, Proposition 10). Since $\mathcal{C}\omega=\chi\omega$, its image lies in the one-dimensional $\mathbb{F}_p$-space

$$L_\chi=\{b\in k: b^p=\chi b\}. \tag{12}$$

By the residue divisor of a differential we mean the formal sum of its residues at geometric points. When compared with an integer divisor, its coefficients are reduced in k, hence modulo p. In particular, logarithmic residues are integer multiplicities reduced modulo p.

Sublemma 3a (Separable pullback). Cartier commutes with pullback by a separable morphism of smooth curves. Indeed, in a separating coordinate x, a rational differential has a unique expression $\sum_{i=0}^{p-1} u_i^p\, x^i\, dx$, and Cartier selects $u_{p-1}\, dx$. A separable pullback keeps the coordinate separating, so applying the same expression after pullback proves the assertion.

For $\chi = 1$, the content of Lemma 3 is classical in substance. The proof follows the kernel differential and quasi-period construction in Voloch (1997): Θ_R in (16) corresponds to his differential $f_P\omega$, and (17) is its translation law. The Cartier invariance of logarithmic differentials is the setting of Gunji (1976) and Serre (1958). We retain a self-contained proof to fix the differential and translation conventions, treat $\chi = -1$ with its factor 2, and obtain the residue-divisor form needed by Theorem 1. The published discussion after Voloch's Proposition 3 prints a negative sign in its quasi-period/logarithmic-derivative formula. With $t_T(Q) = Q + T$ and divisor $p\big((T) - (O)\big)$, the divisor calculation in (15) gives the positive sign used here; the proof therefore determines the sign directly.

Lemma 3 (Exceptional Cartier defect). Suppose $\tau=\chi\in\{1,-1\}$. Let D_0 be a degree-zero integer divisor whose class is killed by p and whose coefficient conjugate is $\sigma(D_0)=\chi D_0$ as divisors. If θ_{D_0} is defined over $\mathbb{F}_{p^2}$, has only simple poles with residue divisor D_0, and satisfies $\sigma(\theta_{D_0})=\chi\theta_{D_0}$, then

$$(\mathcal{C}-1)\theta_{D_0}=\Psi_p([D_0])\omega. \tag{13}$$

Equivalently, the Cartier-defect map and the logarithmic transfer are the same additive map from geometric p-torsion into L_χ; the normalization factor is exactly 1.

Proof. First consider $D_P=(P)-(O)$ for $\chi=1$, or $D_P=(P)-(-P)$ for $\chi=-1$, where $P\neq O$ is geometric p-torsion. Let V be Verschiebung, identifying the Frobenius twist with E through its model over $\mathbb{F}_p$. The Frobenius relations and invariant differential give

$$\pi+V=[\chi], \qquad \pi V=[p], \qquad V^*\omega=\chi\omega, \qquad \mathcal{C}\omega=\chi\omega. \tag{14}$$

The curve is ordinary, so V is separable and étale of degree p. Its kernel $K=\ker V(k)$ is the cyclic group of geometric p-torsion, and every $T\in K$ automatically satisfies $\pi(T)=\chi T$. Thus $K=E(\mathbb{F}_p)$ if $\chi=1$; if $\chi=-1$, it is the group $\{T:\pi(T)=-T\}$ corresponding to the rational points on the anomalous quadratic twist.

The divisor $\Delta=\sum_{T\in K}(T)-p(O)$ is principal: its group sum is O, because nonzero points pair as $T,-T$. It is Galois stable. The one-dimensional Riemann–Roch space $L(-\Delta)$ is defined over $\mathbb{F}_p$, so it has a nonzero base-field element s with $\mathrm{div}(s)=\Delta$. Put $\xi=d\log s$ and $\psi(T)=\Psi_p([(T)-(O)])$. With translation $t_T(Q)=Q+T$, permutation of the kernel points gives

$$\mathrm{div}\,\frac{t_T^* s}{s}=p(O)-p(-T), \qquad t_T^*\xi-\xi=\psi(T)\omega. \tag{15}$$

The sign is positive because $[(O)\text{-}(\text{-}T)]=[(T)\text{-}(O)]$. Also $C\xi=\xi$ and $\psi(T)^p=\chi\psi(T)$.

For $R\in E(k)$, put $Q=V(R)$. Equation (15) makes $t_{\text{-}R}^{*}\xi\text{-}\xi$ invariant under K. Division by $V^{*}\omega$ gives a K-invariant rational function, which descends through the separable quotient V. Hence there is a unique rational differential Θ_R on the target with

$$V^{*}\Theta_R=t_{\text{-}R}^{*}\xi\text{-}\xi=d\log\frac{t_{\text{-}R}^{*}s}{s},$$
$$\operatorname{div}\frac{t_{\text{-}R}^{*}s}{s}=V^{*}\big((Q)\text{-}(O)\big)\text{-}p\big((R)\text{-}(O)\big). \tag{16}$$

As V is étale, Θ_R has only simple poles and residue divisor $(Q)\text{-}(O)$; the terms divisible by p have zero logarithmic residues. Sublemma 3a and injectivity of separable pullback give $C\Theta_R=\Theta_R$. Translation and coefficient conjugation give

$$\Theta_{R+T}=\Theta_R\text{-}\chi\psi(T)\omega \quad (T\in K), \qquad \sigma(\Theta_R)=\Theta_{\pi(R)}. \tag{17}$$

Choose R with $V(R)=P$, so $\pi(R)=\chi R\text{-}P$. For $\chi=1$, set $\alpha=\Theta_R$ and $\beta=\psi(P)$; for $\chi=\text{-}1$, set $\alpha=\Theta_R\text{-}\Theta_{\text{-}R}$ and $\beta=2\psi(P)$. In each case α has residue divisor D_P and $\beta=\Psi_p([D_P])$. The factor 2 is required because $[(P)\text{-}(\text{-}P)]=2P$. Equation (17) gives

$$C\alpha=\alpha, \qquad \sigma(\alpha)=\chi\alpha+\chi\beta\omega, \qquad \beta^p=\chi\beta. \tag{18}$$

In the negative case, explicitly, $\sigma(\alpha)=\Theta_{\text{-}R\text{-}P}\text{-}\Theta_{R+P}=\text{-}\alpha\text{-}2\psi(P)\omega$. A prescribed eigen differential θ_P differs from α by $c\omega$. Its eigencondition gives

$$c^p\text{-}\chi c=\text{-}\chi\beta, \qquad \chi c^{1/p}\text{-}c=\beta. \tag{19}$$

Thus $(C\text{-}1)\theta_P=\beta\omega$.

To pass to D_0, represent its class by D_P, choosing half the class when $\chi=\text{-}1$; multiplication by 2 is invertible on the group of order p. The zero class is handled by $d\log G$ directly. A differential with residues D_P can be chosen over $\mathbb{F}_{p^2}$ and replaced by $(\eta+\chi\sigma\eta)/2$ to satisfy the eigencondition; here $\sigma^2\eta=\eta$. Write $D_0\text{-}D_P=\operatorname{div}(G)$. Its eigencondition implies $G^{\sigma}=\text{constant}\cdot G^{\chi}$, so $d\log G$ is an eigen differential. The difference $\theta_{D_0}\text{-}\theta_P\text{-}d\log G=b\omega$ has $b^p=\chi b$, hence is Cartier fixed. Cartier also fixes $d\log G$. This proves (13).

This reduction shows that the defect depends only on the divisor class: changing an admissible differential adds a Cartier-fixed $b\omega$, and a principal eigen divisor contributes $d\log G$, also Cartier fixed. The map is additive. Its image lies in L_χ, as does the injective transfer (11). The proof determines their comparison scalar as 1, including the factor of 2 in the negative eigencase. The kernel function and the preimage under V are proof constructions and are not computed by the evaluator.

Completion of Theorem 1. In the exceptional case put $\hat{\theta}=\text{-}\theta_\lambda/r$. It has residue divisor D and coefficient eigenvalue χ. Lemma 3 gives $(C\text{-}1)\hat{\theta}=\Psi_p(D)\omega$. Since $r^{1/p}=\chi r$, evaluate at O to obtain

$$S_f(\lambda)^{1/p}=\chi\text{-}\chi r\Psi_p(D), \qquad \Big(\chi r\Psi_p(D)\Big)^p=\chi r\Psi_p(D). \tag{20}$$

Raising to the pth power proves (4) with $H=\chi$.

For concrete exceptional examples, direct expansion and the normalized Miller evaluator give the following values in $\mathbb{F}_{11}$, with $\lambda=1$ and origin $(0,1)$:

$f(w)$	τ	$d=f(1)$	χ	H	$S_f(1)$	$r\Psi_p(D)$
$1+w+3w^3$	1	5	1	1	9	3
$1+w+2w^2+3w^3$	-1	7	-1	10	0	1

Remark on the high part. Put $G(\mu)=\sum_{m\geq p} c_m\,\mu^m$. The Euler-criterion identity after (2) and Theorem 1 give

$$G(\mu)=\chi-H+\chi r\Psi_{M_\chi}(D).$$

In the exceptional case $H = \chi$, this reduces to $G(\mu) = \chi r \Psi_p(D)$. For a trace 1 short Weierstrass curve $y^2 = F(x)$, with origin at infinity and $\omega_0 = dx/(2y)$, the corresponding classical form is

$$F(x)^{(p-1)/2} = \mathcal{U}(x) + x^{p-1} + x^p \mathcal{V}(x), \qquad y\mathcal{V}(x) = \Psi_p([(P) - (O)]), \quad P = (x, y).$$

Here $\deg \mathcal{U} \leq p - 2$; the polynomial $\mathcal{V}$ is distinct from Verschiebung V. Its product with y is the Artin–Schreier descent function, and the right side is the logarithmic transfer (Deuring, 1941; Cassels, 1971; Voloch, 1990, n.d.; Semaev, 1998).

For completeness, fix a nonidentity rational point $P = (a, b)$. Since $E(\mathbb{F}_p)$ has odd order p, $b \neq 0$ and $F(a)^{(p-1)/2} = 1$. The differential $\vartheta_P = (y + b)\omega_0/(x - a)$ has simple poles with residue divisor $(P) - (O)$. Put $q(x) = \big(F(x)^{(p-1)/2} - 1\big)/(x - a)$. Polynomial division gives $[x^{p-1}]q = \mathcal{V}(a)$, and $\deg q < 2p - 1$. Applying Cartier to $dx/(x - a)$ and using $1/y = F(x)^{(p-1)/2}/y^p$ therefore yields

$$(C - 1)\vartheta_P = b\mathcal{V}(a)\omega_0.$$

Lemma 3, with $\chi = 1$ and differential ω_0, gives the displayed transfer identity. At O, both maps are defined to be zero.

Computing the logarithmic derivative

Use the quartic-to-Weierstrass construction with the rational point as origin (Connell, 1999, Section 1.2); the following formulas specify the version and scale required here. Write f=1+f_1w+f_2w^2+f_3w^3+f_4w^4, with f_4=0 for a cubic, and set

$$x=\frac{2(v+1)+f_1w}{w^2}, \qquad y=\frac{4(v+1)+2\left(f_1w+f_2w^2\right)-f_1^2w^2/2}{w^3}. \tag{21}$$

Substitution gives the generalized Weierstrass equation with

$$a_1=f_1, \quad a_2=f_2-f_1^2/4, \quad a_3=2f_3, \quad a_4=-4f_4, \quad a_6=a_2a_4. \tag{22}$$

Near P_+, direct expansion gives

$$\begin{aligned} x&=4w^{-2}+2f_1w^{-1}+f_2-f_1^2/4+O(w),\\ y&=8w^{-3}+4f_1w^{-2}+(4f_2-f_1^2)w^{-1}+O(1). \end{aligned} \tag{23}$$

Thus P_+ maps to the identity at infinity. The points at infinity on the quartic map to affine points on this model; the finite points $Q_\pm$ also have affine images $R_\pm \neq O$, since $1/\lambda \neq 0$.

Put $b_2=a_1^2+4a_2$, $b_4=2a_4+a_1a_3$, $b_6=a_3^2+4a_6$, and use $X=x+b_2/12$, $Y=y+(a_1x+a_3)/2$. The short equation is $Y^2=X^3+AX+B$, where

$$A=b_4/2-b_2^2/48, \qquad B=b_6/4-b_2b_4/24+b_2^3/864. \tag{24}$$

All denominators are units for $p \geq 7$. The invariant differentials are proportional; comparing their leading terms in (23) fixes

$$\omega=\kappa\frac{dX}{2Y}, \qquad \kappa=-2. \tag{25}$$

For a point R on the short model, use Miller functions and their local constants

$$\begin{aligned} \operatorname{div}\left(f_{m,R}\right)&=m(R)-([m]R)-(m-1)(O),\\ \alpha_m(R)&=[z^0]\frac{d\log f_{m,R}}{dz}, \quad z=-X/Y. \end{aligned} \tag{26}$$

This is a Laurent coefficient of df/f, not an analytic logarithm, and multiplication of a Miller function by a scalar does not change it. The divisor recurrence is standard (Miller, 2004; Galbraith, 2012, Section 26.3.1). A nonvertical chord or tangent of slope ρ and a vertical line have expansions

$$\begin{aligned} Y-\rho X-v&=-z^{-3}\left(1+\rho z+O(z^3)\right),\\ X-c&=z^{-2}\left(1-cz^2+O(z^4)\right). \end{aligned} \tag{27}$$

The constant logarithmic derivative is therefore $+\rho$ for the nonvertical line and 0 for the vertical line. Binary doubling and addition update the accumulator by

$$\alpha \leftarrow 2\alpha+\rho, \qquad \alpha \leftarrow \alpha+\rho, \tag{28}$$

respectively. Vertical and identity steps contribute zero. Doubling the accumulator is still necessary when the current point is O.

Since $[M_\chi]R_+=[M_\chi]R_-$, the quotient $f_{M_\chi,R_+}/f_{M_\chi,R_-}$ has divisor $M_\chi D$. Its pole terms at O cancel, and $dX/(2Y)=\left(1+O(z^4)\right)dz$, so

$$\Psi_{M_\chi}(D)=\frac{\alpha_{M_\chi}(R_+)-\alpha_{M_\chi}(R_-)}{\kappa}. \tag{29}$$

Compute without finding a square root in the two-dimensional étale algebra $\mathcal{A}_d=\mathbb{F}_p[s]/(s^2-d)$. Run the chain with $v_0=s$ and write its constant as $\alpha=A_0+B_0 s$. Algebra conjugation gives the other chain. Equations (4) and (29) become

$$r\Psi_{M_\chi}(D)=\frac{2B_0}{\kappa\lambda}=-\frac{B_0}{\lambda}, \qquad S_f(\lambda)=H+\chi\frac{B_0}{\lambda}. \tag{30}$$

For $\chi=-1$, the algebra is a field; for $\chi=1$, it is a product of two fields. The latter requires branch tests componentwise. An element $u=A_0+B_0 s$ with norm zero and $u\neq 0$ has $B_0\neq 0$ and supplies the root $(A_0/B_0)^2=d$. Split the saved state by $s\mapsto\pm A_0/B_0$ and resume the remaining binary steps in both fields. At addition, first handle an identity point; for equal abscissae use a vertical step when ordinates are opposite, and a tangent when the points coincide, treating an ordinate of zero as vertical. Otherwise use the chord. At doubling, an identity or ordinate-zero point is a zero-contribution step; otherwise use the tangent. A nonzero zero divisor in an equality test can also split the state before choosing these branches, so a tangent in one component and a vertical addition in the other are handled separately. Splitting can occur only once. This is the degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985).

Algorithm for the differentiated Miller chain. The following pseudocode uses the short model and computes the constant in (26). Prepare the operation stream from the binary expansion of M, omitting its leading 1: each remaining bit contributes a doubling operation followed by an addition operation if the bit is 1. The stored operation index is the next uncommitted operation.

```
T := R; alpha := 0; j := 0
while j < length(operations):
    op := operations[j]
    Q := T if op is doubling, otherwise R
    attempt (Tnew, rho) := ADD(T, Q)
    if a nonzero zero divisor a + b*s is found:
        split saved R, T, alpha by s -> +a/b and -a/b
        resume both field components at this same j; return both constants
    anew := 2*alpha + rho if doubling, otherwise alpha + rho
    (T, alpha, j) := (Tnew, anew, j + 1)
return alpha, T
```

The subroutine ADD returns $(Q,0)$ if $T=O$, and $(T,0)$ if $Q=O$. For $T=(X_1,Y_1)$ and $Q=(X_2,Y_2)$, it returns $(O,0)$ if $X_1=X_2$ and $Y_1+Y_2=0$; this includes doubling a point of order 2. If $T=Q$ otherwise, use the tangent slope $(3X_1^2+A)/(2Y_1)$; if $X_1\neq X_2$, use the chord slope $(Y_2-Y_1)/(X_2-X_1)$. Then set $X_3=\rho^2-X_1-X_2$ and $Y_3=\rho(X_1-X_3)-Y_1$, and return $\left((X_3,Y_3),\rho\right)$. If $X_1=X_2$ but the tangent and vertical tests disagree between components, Y_1-Y_2 supplies the splitting witness. Any attempted inverse of a nonzero element of norm zero also supplies a witness. Its coefficient b is nonzero and $(a/b)^2=d$. The split uses the saved state before the failed operation, so no accumulator update is repeated. In either component, identity and vertical steps still apply the doubling update when required. Finally use (29), or (30) before a split, to recover the coefficient sum.

For fixed f and p, cache the model, exact trace, H, and $M_\pm$. Character computation and the affine Miller chain use $O(\log p)$ base-field multiplications and inversions. Resuming at most two components preserves this bound. The streaming query needs a constant number of working field elements and $O(\log p)$ control bits; an implementation retaining chain states may instead use $O(\log p)$ field elements.

Corollary 4 (Deterministic polynomial-time evaluation). Given $p\geq 7$, a squarefree cubic or quartic $f\in\mathbb{F}_p[w]$ with $f(0)\neq 0$, and an admissible λ, $S_f(\lambda)$ is computable deterministically in time polynomial in $\log p$. For N_q queries on the prepared curve, classical arithmetic gives

$$T_{\text{total}}=T_{\text{Schoof}}(p)+O\big(N_q(\log p)^3\big), \qquad T_{\text{Schoof}}(p)=O((\log p)^9). \tag{31}$$

The stated setup bound is the conservative bound of Schoof's original algorithm, not an optimal point-counting bound. After setup, a query uses $O(\log p)$ field operations and at most $O(\log p)$ field elements plus $O(\log p)$ control bits. This query-space bound does not include the polynomial working space of point counting.

Proof. If $c_0=f(0)\neq 1$, normalize $g=f/c_0$ and restore $S_f=c_0^h S_g$. Use the trace of $v^2=g(w)$, which is a quadratic twist of the original curve when c_0 is nonsquare. Fixed-degree squarefreeness tests and the model change are polynomial in $\log p$. Schoof's deterministic point count gives the trace and annihilators (Schoof, 1985; see Schoof, 1995 for the survey). Theorem 1 includes exceptional traces. All query algebra has fixed dimension; a zero divisor supplies its own splitting root. Since $M_\chi=O(p)$, the chains have $O(\log p)$ length. Classical modular multiplication and inversion cost $O((\log p)^2)$ bit operations, giving (31). At $\lambda=0$ use the trace residue of the normalized curve and restore c_0^h.

Practical trace preparation can use SEA or an automatic exact routine such as PARI/GP's `ellap`, with its available SEA data at large characteristic (PARI Group, n.d.). The exact trace can be cached once per curve and prime; a change of curve may require a new trace. These practical choices do not replace Schoof in the deterministic worst-case proof. The appendix uses specialized Cornacchia/Gauss preparation and does not benchmark a Schoof-based implementation of this corollary. For many queries, computing and sharing a full coefficient row can still be preferable.

Application to the original quarter point sum

For an odd prime p, put $h=(p-1)/2$ and define the rational sequence

$$t_0=1, \qquad t_{i+1}=t_i\,\frac{h+2+i}{2i+2}, \qquad U_p(L)=\textstyle\sum_{i=0}^{L-1}(h+1+i)\,t_i. \tag{32}$$

Reduction is well defined for $0\leq L\leq h$, because the denominators use powers of two and factorials below p. Put $a_i=\binom{2i}{i}8^{-i}$ and $B_L=\sum_{i<L}a_i$. Upper negation gives, in $\mathbb{F}_p$,

$$t_i=2^{-i}\binom{h+1+i}{i}\equiv 2^{-i}(-1)^i\binom{-3/2}{i}=(2i+1)\binom{2i}{i}8^{-i}. \tag{33}$$

The ratio $a_{i+1}/a_i=(2i+1)/\big(4(i+1)\big)$ gives the telescoping certificate

$$((2i+1)^2-8)a_i=-4(i+1)(2i+7)a_{i+1}+4i(2i+5)a_i. \tag{34}$$

Consequently,

$$U_p(L)\equiv 4B_L-2L(2L+5)a_L \pmod p. \tag{35}$$

Both the prefix and the boundary coefficient are required for this weighted sum. Now take $p=4n+1\geq 13$, so $h=2n$. Since $\binom{h}{i}(-1/2)^i=a_i$ in $\mathbb{F}_p$, B_n is a strict midpoint truncation of the binomial row of length $h+1$, although $n=(p-1)/4$ is a quarter point relative to p. The full row sum is $(1/2)^h=(2/p)$.

Set $f(w)=1-w^4/2$. Its coefficient of w^{4i} in f^h is a_i, so $H=a_n$ and $S_f(1)=B_n+a_n$. Theorem 1 and (35) give

$$B_n=-\chi r\Psi_{M_\chi}(D), \qquad U_p(n)\equiv 4B_n+\frac{9}{4}a_n \pmod p, \tag{36}$$

where $\chi=(2/p)$ and $r=1/v_0$, $v_0^2=1/2$. The curve has $j=1728$ and rational nonzero two-torsion, so its trace is even and the exceptional traces ± 1 do not occur in this particular application.

The trace is classical complex-multiplication data. With $p=u^2+q^2$ and the odd coordinate normalized by $u\equiv 1 \pmod 4$, Gauss's congruence is

$$\binom{2n}{n}\equiv 2u \pmod p, \qquad a_n\equiv 2u\,8^{-n} \pmod p. \tag{37}$$

This convention agrees with Cosgrave and Dilcher (2010, Theorem 1). The trace is the unique even integer in the Hasse interval with residue a_n. If $\bar{a}_n \in \{0,\ldots,p\text{-}1\}$ is its least nonnegative residue, it is $\bar{a}_n$ when that residue is even, and $\bar{a}_n\text{-}p$ otherwise. Cornacchia preparation and exponentiation therefore supply the boundary and trace for a Miller evaluation of the prefix. The benchmarked preparation includes a sequential nonresidue scan, for which no unconditional polylogarithmic worst-case bound is claimed. Corollary 4 provides an unconditional deterministic polynomial-time alternative: compute the curve's exact trace, use a_n=H, and evaluate B_n=$S_f(1)$-H. At p=97, n=24, B_n=33, a_n=79 and $U_p(n)$=43.

The congruence literature for truncated central-binomial sums includes Sun (2011). The example here makes no separate priority claim for its binomial consequences. The elliptic-Fermat-quotient interpretation is established in the anomalous setting of Satoh and Araki (1998, 1999); applying that terminology to the regular $\Psi_{M_\chi}(D)$ in this example is only an analogy, since it retains a specified divisor and annihilator.

Rounded third prefix and boundary

For primes $p \geq 11$ with $p \equiv 2 \pmod 3$, set $m = (p-2)/3$ and $L = m + 1 = (p+1)/3$. Retain the definitions of a_i, the strict prefix B_L and $U_p(L)$ from (32)–(35). Here the prefix is another direct application of Theorem 1: take $f(w) = 1 - w^3/2$ and $\lambda = 1$. The coefficients of f^h at w^{3i} are a_i, and $3m < p \leq 3L$. Thus $S_f(1) = B_L$. The coefficient H vanishes because $3 \nmid p-1$; the exact trace is zero by the trace congruence and Hasse's bound. Since $f(1) = 1/2 \neq 0$, the argument is admissible, and the differentiated Miller query evaluates B_L in $O(\log p)$ field operations, without a point-counting search in this family.

The boundary a_L needed in (35) has a different arithmetic interpretation. Put $G = m! \bmod p$. Wilson reflection and the ratio of consecutive terms give

$$a_L = \frac{3}{2G^3}, \qquad G = \left(\frac{3}{2a_L}\right)^{(2p-1)/3} \quad \text{in } \mathbb{F}_p. \tag{RT1}$$

Indeed, $(2m)!\,L! = -1$ and $L! = LG$ give $(2m)! = -3G^{-1}$. Also $8^m = 1/2$, so $a_m = -6G^{-3}$, while $a_L/a_m = -1/4$. Cubing is bijective on $\mathbb{F}_p^\times$, yielding the inverse formula. Thus either a boundary backend or a modular-factorial backend supplies the other with $O(\log p)$ additional field operations. This equivalence is not a computational lower bound.

The cubic above becomes $E: y^2 = x^3 + 1/4$ under $x = -w/2$, $y = v/2$. For the following cohomological statement use the point at infinity O as origin and fix $\omega = dx/y$; this differential has the opposite sign to dw/v under that coordinate change. Write $F = x^3 + 1/4$ and $F^h = T + x^p S$, with $\deg T < p$. Let U be E with O removed, and W be E with $(0,1/2)$ and $(0,-1/2)$ removed. A Čech–de Rham cocycle is (α_U, α_W, g) with $\alpha_U - \alpha_W = dg$, and a coboundary is $(du_U, du_W, u_U - u_W)$. Fix

$$e_1 = [(\omega, \omega, 0)], \qquad e_2 = \left[\left(x\omega, \frac{\omega}{2x^2}, \frac{2y}{x}\right)\right]. \tag{RT2}$$

Proposition 5 (Normalized rounded third boundary). The classes (e_1, e_2) form a basis of $H^1_{\mathrm{dR}}(E/\mathbb{F}_p)$. With the Cartier convention of (6),

$$d(yS) = a_L\omega, \qquad \mathcal{C}(\omega) = 0, \qquad \mathcal{C}(x\omega) = \mathcal{C}\left(\frac{\omega}{2x^2}\right) = 2a_L\omega. \tag{RT3}$$

Define $V_{\mathcal{C}}[(\alpha_U, \alpha_W, g)] = [(\mathcal{C}\alpha_U, \mathcal{C}\alpha_W, 0)]$ and $F_{\mathrm{abs}}[(\alpha_U, \alpha_W, g)] = [(0,0,g^p)]$. These maps are linear over $\mathbb{F}_p$. With columns giving images of basis vectors,

$$[V_{\mathcal{C}}] = \begin{pmatrix} 0 & 2a_L \\ 0 & 0 \end{pmatrix}, \qquad [F_{\mathrm{abs}}] = \begin{pmatrix} 0 & -2a_L \\ 0 & 0 \end{pmatrix}. \tag{RT4}$$

Proof. For a polynomial P, put $\Theta(P) = FP' + F'P/2$, so $d(yP) = \Theta(P)\omega$. Since $h + 1/2 = 0$ in the field, $\Theta(T) = -x^p\Theta(S)$. The expansion of F^h has exponents divisible by three, with $\deg T = p-2$, so $\deg\Theta(T) \leq p$. Consequently $\Theta(S)$ is constant, equal to $[x^{p+1}]F^h/4 = a_L$. Cartier coefficient selection gives $[x^{p-1}]F^h = 0$, $[x^{p-2}]F^h = 2a_L$ and $[x^{p+1}]F^h = 4a_L$, proving (RT3).

The identity $d(2y/x) = \left(x - 1/(2x^2)\right)\omega$ verifies the second cocycle. Its transition function has a simple pole at O. A Čech coboundary would require a function regular away from O with just that simple pole, excluded

by genus-one Riemann–Roch. Together with the nonzero regular differential class, it therefore gives a basis. Applying Cartier proves the first matrix. For the second, write $(2y/x)^p = u_U - u_W$ with $u_U = 2yS$ and $u_W = -2yT/x^p$. These functions are regular on their respective opens, and $du_U = du_W = 2a_L\omega$. Subtracting their coboundary from $(0,0,(2y/x)^p)$ gives $-2a_L e_1$. This proves (RT4). The companion extension supplies the regularity and basis arguments in full and states the Frobenius-twist conventions over perfect extensions (Oda, 1969).

The usual genus-one Cartier–Manin matrix on regular differentials is the single entry zero. The nonzero entry $2a_L = 3G^{-3}$ in (RT4) belongs to full de Rham cohomology in this specified basis. Although $a_L\omega$ is exact in the function field, it is nonzero in projective de Rham cohomology: its rational primitive has poles and cannot be made globally regular. This is a global regularity obstruction, not a prohibition on cancelling a pole at one point. Neither the zero trace nor this interpretation provides a fast boundary extractor, and no identification with a normalized Chowla–Selberg period is used.

Two checked computational routes are supplied in the companion extension. For dense prime ranges, an accumulating remainder forest computes $\left(B_L, a_L, U_p(L), G\right)$ directly from an integer triangular recurrence, sharing work across primes. This specializes established remainder-forest machinery (Harvey & Sutherland, 2016) and does not improve isolated-prime complexity. For one prime, small-conductor Jacobi sums or trace-one character sums supply a nearby a_k, and two interval products correct from k to L using the consecutive-term ratio. The anchor constructions follow the Jacobi-sum literature and the unrefereed Tal direction (van Wamelen, 2002; Tal, 2026); the bounded implementation supplies its own normalization and reconstruction certificates, without assuming Tal's broader complexity theorem.

With conductor at most 90 and cyclotomic degree at most 24, the implemented anchor route retains the square-root exponent. It improved complete evaluation times by factors of 1.37–4.82 against FLINT 3.6.0's native modular-factorial routine in five selected 28–40-bit examples; all 30 timed output tuples agreed. These illustrative measurements include boundary reconstruction and use the same Miller prefix in both methods. They establish neither a population-wide gain nor superiority over SageMath or Magma. Smaller inputs lost time in the separate anchor experiment. The companion reports the timing exclusions, selected inputs, fallbacks and raw repetitions. Its rounded-third comparisons are separate from the quarter-point pilot in Table 1.

The uniform polylogarithmic single-prime boundary question remains open. Theorem 1 and Corollary 4 concern their original admissible coefficient-sum problem; they do not imply such a bound for this additional boundary or for the complete rounded-third weighted sum.

Scope and an independent point sum

At a branch argument $f(1/\lambda)$=0, the two simple poles of θ_λ merge into a residue-free pole of the second kind. Removing exact differentials leaves a second-kind cohomology class whose Cartier image involves an additional polynomial-power coefficient, beyond the trace. Katz's note (n.d.) illustrates this distinction through Cartier on dx/y and $x\,dx/y$. This observation does not establish a fast branch evaluator. The current division-free streaming fallback takes $O(p)$ field operations.

If $f(0)$=0, squarefreeness gives f=wg with $g(0)\neq 0$, and the coefficient problem becomes $[w^h]g(w)^h/(1\text{-}\lambda w)$ rather than the normalized cutoff p-1. The origin and local coefficient selection used above therefore change; this case is not included in Theorem 1.

Proposition 2 already holds over perfect extension fields. For q=p^k and $\lambda\in\mathbb{F}_q^\times$, put ε=$d^{(q\text{-}1)/2}\in\{1,\text{-}1\}$ and let τ_k be determined by τ_0=2, τ_1=τ, τ_k=$\tau\tau_{k\text{-}1}$-$p\tau_{k\text{-}2}$. The divisor class is annihilated by M=q+1-$\varepsilon\tau_k$. If $p\nmid M$, (9) gives $S_f(\lambda)$=H+$(r/M)(H\Psi_M(D)$-$\Psi_M(D)^p)$, with a Miller chain of length $O(k\log p)$. Here $p|M$ exactly when $\varepsilon\tau^k\equiv 1 \pmod p$. The simplified base-field formula and the equality with $T_f(1/\lambda)$ do not automatically extend, and no uniform exceptional extension-field theorem or implementation is claimed.

An elementary verification identity gives an independent $O(p)$ computation for admissible base-field arguments. With μ=$1/\lambda$ and d=$f(\mu)$,

$$S_f(\lambda)=\chi\text{-}\frac{\chi f'(\mu)}{2d\lambda}\text{-}\sum_{\substack{P\in E(\mathbb{F}_p)\ \text{affine}\\ w(P)\neq\mu}}\frac{1}{1\text{-}\lambda w(P)}. \tag{38}$$

To check it, write $F=f^h$ and $Q=\big(F\text{-}F(\mu)\big)/(w\text{-}\mu)$. The number of affine points over w is $1+F(w)$. Summing after polynomial division, using (8) and $Q(\mu)=F'(\mu)$, yields the point sum $F(\mu)\text{-}T_f(\mu)+F'(\mu)/\lambda$. Since $F'(\mu)=\text{-}\chi f'(\mu)/(2d)$, this is (38). Points at infinity contribute zero if the summand is extended by its limiting value. This identity neither uses Miller functions nor proves their exceptional normalization by itself.

Since $\sum_{w\neq\mu}(1-\lambda w)^{-1}=0$ in $\mathbb{F}_p$, counting the $1+\chi\big(f(w)\big)$ affine points above each w rewrites (38) as the character sum

$$S_f(\lambda)=\chi-\frac{\chi f'(\mu)}{2d\lambda}+\mu\sum_{w\neq\mu}\frac{\varepsilon(w)}{w-\mu}.$$

Here $\varepsilon(w)=(f(w)/p)$ is the quadratic character, with $\varepsilon(w)=0$ when $f(w)=0$; the scalar χ still denotes the character of d. Thus the evaluator also computes this rationally weighted character sum without enumerating the base field.

Computational verification

Retained code compares the generic evaluator with direct polynomial powers and independent point counts. An audit over primes $11\leq p\leq 499$ tested 20,316 admissible queries on 906 sampled squarefree cubics and quartics, including 368 exceptional queries and 2,512 split-algebra fallbacks, with no mismatches. The attribution revision reran this sample and additionally checked the high-part identity in the remark after Theorem 1 on every query; the character-sum form following (38) agreed on its 4,072 admissible queries with $p<90$. Exhaustive compiled tests over $\mathbb{F}_5$ and $\mathbb{F}_7$ agreed with FLINT polynomial powers (Hart, 2010); the $p=5$ checks are computational evidence outside the stated theorem.

A separate exhaustive check enumerated all 343 nonsingular short Weierstrass models $y^2=x^3+Ax+B$ of trace 1 for $7\leq p<48$. At all 11,730 nonidentity rational points, direct expansion of the high part agreed with the independent Miller slope accumulator relative to $dx/(2y)$. The identity point was checked separately on each model, and all 463,615 ordered pairs of rational points satisfied additivity. The supplementary $p=5$ check covered two models and eight nonidentity points, outside the main theorem's prime range.

Separate retained direct-proof checks covered 17,940 nonzero torsion-point evaluations on 354 curves for $7\leq p\leq 101$, and 568 kernel-translation checks on 32 curves for $7\leq p\leq 31$. An independent point-sum verifier tested 4,357 queries on 384 curves for $7\leq p\leq 103$, including 155 exceptional queries, with no mismatches. Source code, detailed logs and reproduction commands accompany the manuscript and the rounded-third companion extension. These finite checks do not replace proof review.

The rounded-third extension has a separate validation record. The anchor suite checked complete outputs against the original definitions at all 152 branch primes from 11 to below 2,000, and 595 reconstructed conductor values against direct character enumeration. Its main comparison comprised 264 fresh-process evaluations over 22 primes, two methods and two product backends, with three repetitions. A separate direct native-FLINT comparison comprised 30 complete evaluations over five selected primes. The standard-library batch produced 39,264 tuples below one million; independent comparisons covered every prefix, all 1,135 complete tuples below 20,000, and 40 reproducible larger complete tuples. These scopes are different and are recorded separately. The unshared table below 100 million is not included in the independently verified evidence. The companion retains proofs, source hashes, raw measurements and reproduction commands; finite agreement does not replace proof review.

Further questions

The identity concerns admissible genus-one inputs with a fixed coefficient cutoff. Projective arithmetic and a practical exact point-counting frontend could improve its implementation. Second-kind branch arguments, exceptional extension-field cases and other stopping families merit separate treatment. A higher-modulus congruence would require new arguments: the characteristic-p Cartier proof does not establish evaluation modulo p^2. For trace 1 the normalization is classical in substance (Deuring, 1941; Cassels, 1971; Gunji, 1976;

Voloch, 1990, 1997, n.d.). We have not located the trace -1 residue-divisor statement or the two-pole base-field form in these sources. Specialist comparison is needed before making a priority claim for either formulation.

For the rounded-third family, the remaining question is a uniform polylogarithmic single-prime extractor for the normalized scalar in (RT4), equivalently the Gauss factorial in (RT1). The checked batch route amortizes across primes, while the bounded Jacobi-anchor route can reduce the remaining interval at one prime. Neither the fixed-conductor implementation nor the de Rham interpretation resolves this asymptotic question. Cost-aware selection between anchor and factorial backends, additional platform validation and independent specialist review are practical next steps. No general computational lower bound is claimed.

AI assistance

Generative AI was used extensively in mathematical exploration, programming, computational checking and manuscript preparation under the author's direction. The reported computational checks arose within that research programme and the supplied AI feedback; they are not human peer review. The manuscript remains unrefereed by a human specialist.

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Appendix: Pilot Benchmark

Table 1 presents a quarter-point pilot for seven primes $p \equiv 1 \pmod 4$, with $n=(p-1)/4$. Both implementations compute B_n, a_n and $U_p(n)$, including per-input preparation. The comparator is a standalone C++ adapter to the unchanged SageMath 10.8 Harvey `hypellfrob` interval-product kernel, which uses BGS-based recurrence techniques (Bostan et al., 2007; Harvey, 2007; SageMath Developers, n.d.). This is a benchmark of that recurrence task, not of Harvey's full Kedlaya algorithm.

Table 1. Complete quarter-point evaluation in milliseconds: Python Cartier–Miller versus the C++ Harvey/BGS recurrence adapter, including per-input preparation

Prime p	Stopping index n	Cartier–Miller Python	Harvey BGS C++
97	24	0.05	0.01
1,009	252	0.06	0.05
10,009	2,502	0.07	0.10
100,049	25,012	0.09	0.54
1,000,033	250,008	0.10	1.78
10,000,121	2,500,030	0.11	4.44
100,000,037	25,000,009	0.29	16.19

The run used a shared hosted Linux environment reporting an AMD EPYC 9V74 processor, Python 3.12.14, GNU C++ 13.3.0 with `-O3`, and NTL 11.5.1 configured for one thread. Times are medians of 11 per-call timing-batch means, with alternating method order in warmed persistent processes. Batches targeted about 25 ms and were capped at 2,000 evaluations. Answers and per-call mathematical data were recomputed; library tables could remain warm. Validation, compilation, primality testing, process startup, communication and output writing were excluded. Host load and effective CPU frequency were not controlled.

All timed outputs agreed. Adapter validation additionally covered 10,809 stopping-point queries across 92 primes from 7 to 499, and 328 quarter-point comparisons for primes $p \equiv 1 \pmod 4$ from 13 to 4,999, with no mismatches. The timing run did not change these validation counts.

Cornacchia/Gauss preparation includes a nonresidue scan; this pilot does not implement Schoof setup or establish Corollary 4's worst-case bound. The direct characteristic-p proof did not change the evaluator. Small-input entries include call and loop overhead. Seven selected inputs, different implementation languages, an uncontrolled host and unmeasured memory limit general speed or scaling claims. Raw samples and reproduction commands accompany the manuscript.